Final Report for e-Portfolio including reflective

(Research Methods and Professional Practice)

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Word Count: 2198 words

GitHub Repository URL:

https://alahsshmi.github.io/E-Portfolio/Research-Methods.html

Date: 25 Jan 2026

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e-Portfolio Digital Report

Purpose and e-portfolio scope

This piece reviews my learning path through the Research Methods and Professional Practice module. Each class covered core topics in research design, data work, ethics, and career skills. Hands on tasks and group talks backed them all. These steps built my thinking skills and work skills. Next, I outline key lessons from each module part. I draw from proof in my e-portfolio.

It gathers:
- reflective tasks finished in the module
- statistics tasks, with required worksheets from Units 8 and 9
- a review of my Literature Review and Research Proposal
- proof of skills growth through a skills matrix and action plan

Evidence map

The table below lists where each need is met. It shows the proof supplied. Screenshots, figures, spreadsheets, and appendices serve as items. They stay out of the word count.

Table 1. Evidence map

Requirement from brief	Where it is met	Evidence (appendix/figure)
Reflective activities across the module	Digital Report: Unit learning journey (Units 1–5) and reflective activity summaries: Reflective Piece: WHAT–SO WHAT–NOW WHAT sections.	Reflective Activity a Generative AI Impact in Software Security in Emails Case Study: Inappropriate
Statistics exercises incl. Units 8 and 9 worksheets	Digital Report: Statistics Exercises section (Units 6–9 learning) with interpretation of outputs.	Hypothesis Testing Worksheet – Research Version and Summary Measures Worksheet – Research Version
Evaluation of Literature Review submission	Digital Report: Evaluation section (Literature Review) focusing on scope, source strategy, gaps, and planned improvements.	Literature Review and Research Proposal Outlines + Literature Review Guide Questions – Answers
Evaluation of Research Proposal submission	Digital Report: Evaluation section (Research Proposal) covering aims, methodology, artefact, ethics, and improvements.	Research Proposal Presentation Transcript slides 2–10 Research Proposal Review
Skills matrix and SWOT/action plan	Digital Report: Professional Development section linking skill growth to artefacts and a forward plan.	Tables 2 and 3 in this report concepts applied (risk and change control learning).

Learning journey by unit (Units 1-12)

This part sums up my path through the module. For each unit, I note my outputs. I share key lessons. I point to proof in the e-portfolio.

Unit 1: Introduction to Research Methods and Ethics in Computing

I sorted out inductive and deductive reasoning. I tied ethics to daily tech choices. Reflective Activity A on AI email security cast privacy, blame, and harm as design limits (Walliman, 2018; Floridi, 2019; Mittelstadt, 2019).

Unit 2: Research Questions, Literature Review, and Proposal

I shaped a wide topic into clear questions. I built a firm review plan. Standards-based sources kept smart grids and SCADA in real world bounds. This set up the later proposal (NIST, 2015; IEC, 2018).

Unit 3: Methodology and Research Methods

I learned to back method picks with reasons, not old habits. I matched a step-by-step, item focused plan to checks needed. It kept aims, steps, and proof in line (Creswell and Creswell, 2018; Saunders, Lewis and Thornhill, 2019).

Unit 4: Case Studies, Focus Groups, and Observations

I weighed case studies, focus groups, and observation for solid insights. This sharpened my view of context rich systems. Stakeholder habits and limits guide what we can test.

Unit 5: Interviews, Surveys, and Questionnaire Design

I built a sharper eye for tools and sampling. The survey flaw case drove it home: bad questions or weak consent kill trust and rightness. Even if results seem fine.

Unit 6: Quantitative Methods - Descriptive and Inferential Statistics

I honed basic stats for description. I checked spread and shape first before claims. This aided clean reports in my summary sheet and later checks (Saunders, Lewis and Thornhill, 2019).

Unit 7: Inferential Statistics and Hypothesis Testing

I worked on stating guesses, picking tests, and reading results without overreach. The guess test sheet-built care for base ideas, rules, and clear reports (Creswell and Creswell, 2018).

Unit 8: Data Analysis and Visualisation

I saw charts as proof tools, not extras. I picked visuals to fit data and readers. This made portfolio evidence easy to scan (Walliman, 2018).

Unit 9: Validity and Generalisability in Research

I used trust and steadiness ideas to judge sources and my plans. It honed my review checks. It set what proof works in real use.

Unit 10: Research Writing

I boosted layout, guides, and Harvard cites in my items. I tied claims to proof. I kept the story checkable for pros (Walliman, 2018).

Unit 11: Going Forward - Professional Development and e-Portfolio

I turned thoughts to steps via skills grid, SWOT, and PDP. The e-portfolio as sorted record proved growth. It set aimed fixes.

Unit 12: Project Management and Risk

I brought risk views to project flow with a basic risk grid and change log. This aided sound timelines and backups for the proposal (Saunders, Lewis and Thornhill, 2019).

Statistics exercises summary

Stats exercises built three key skills. They turn real scenarios into testable hypotheses. Users pick suitable tests. They list assumptions and interpret results without overclaims.

Worksheets follow one clear format. It covers hypotheses, test choice, decision rule, results, and brief context linked meaning.

For steady reports, I stuck to this format in all tasks. I will use it ahead too:

- Pose the question in one line.
- Set null and alternate hypotheses.
- Test assumptions; flag limits.
- Give decision plus real world meaning.

This habit aids solo work. It shows my thought process. It cuts confusion. Tutors or peers can trace each step. No guesswork on my aims. In time, it splits right maths from trusted analysis.

Evaluation of Literature Review submission

My literature sets a tight scope. It uses a planned search tied to IoT risks in power grids.

What went well:

- Sharp focus on operational tech limits.
- Solid standards and guides with peer reviewed papers.
- Theme links for architecture, threats, and control options.

What to fix next time

- Turn evidence gaps into clear test ideas.
- Add quick quality checks for each source.
- Make a short table (threat, layer, effect, control, proof strength).

Evaluation of Research Proposal submission

Proposal shines in linking study design to utility realities. It states the issue. It lists clear questions. It offers an artefact path open to checks.

Strengths:

- Issue tied to real operations blocks.
- Questions match outputs and checks.
- Firm artefact idea (safety toolkit, testbed setup).

Areas to build:

- Spell out check measures (like ease scores, ops effects).
- Add ethics and rules details (log care, access limits, keep times, agreement).

Professional development

Module boosted job skills via set self-checks. Skills grid compared start skills to now. Each shift links to proof.

I also used a Short SWOT and action list set close goals. They stay trackable and proof based.

Table 2. Skills matrix

Skill area	Before this module	After this module	Evidence
Research question design	I could state a topic interest, but my questions were broad and not always linked to measurable outcomes.	I frame focused research questions that map to aims, objectives, and evaluation, with clear OT constraints in mind.	Research Proposal Presentation Transcript. Slide 4 (RQs) + Slide 3 (aim/objectives).
Literature review planning	I searched for sources, but I did not use consistent inclusion criteria or a clear synthesis framework.	I use databases, keywords, and time bounds, then organise findings with a synthesis framework tied to threats and controls.	Literature Review Outlines + Guide Questions – Answers.
Method justification	I tended to pick methods by familiarity, with limited explanation of why they fit the problem context.	I justify a staged design (Design Science + targeted qualitative feasibility input) aligned to deliverables and evaluation needs.	Research Proposal Presentation Transcript. Slide eight: Research Proposal Review.
Quantitative analysis	I understood descriptive statistics, but I lacked confidence selecting and interpreting hypothesis tests.	I can state hypotheses, choose suitable tests (paired/independent/one tailed), interpret p-values, and report decisions clearly.	Hypothesis Testing Worksheet – Research Version: Summary Measures Worksheet – Research Version.

Research writing	My writing was sometimes descriptive, and I needed stronger signposting and tighter argument flow.	I write in structured sections (problem–method–ethics–evaluation) and support claims with consistent Harvard citations.	Research Proposal Presentation Transcript + reflective writing samples.
Ethics in computing	I treated ethics as a checklist and focused on technical controls rather than governance and user impact.	I critically consider consent, transparency, accountability, and professional duties alongside security measures.	Reflective Activity A Emails: Inappropriate Use of Surveys case analysis.

Table 3. SWOT and action plan

Area	**Point**	**Action I will take**	**Timeline**
Strength	Strong OT/SCADA context awareness and ability to translate security ideas into feasible controls.	Use feasibility criteria (uptime, patch windows, legacy limits) to prioritise controls and justify decisions.	Next proposal iteration (Week 1–2)
Weakness	Evaluation metrics and measurement plan can be tightened to avoid vague conclusions.	Define 4–6 metrics (e.g., latency impact, detection quality, false positives, feasibility score) and link each to a research question.	Within 1 week
Opportunity	Security toolkit can become a reusable professional artefact aligned to NIST/IEC guidance.	Package templates, checklists, and testbed steps as a versioned toolkit with a short user guide and evidence examples.	2–3 weeks
Threat	Limited access to practitioners and the risk of scope creep or testbed complexity affecting timelines.	Keep interviews; rely on simulated data: apply change control and a small risk register to protect scope.	Ongoing (review weekly)

Professional Development Plan (PDP)

This Plan shifts self-check to firm steps. Each tie to proof for e-portfolio. Use screen grabs, extras, or fresh items.

Objective	Key actions	Evidence / success criteria	Support / resources	Target date
Strengthen evaluation design for the research proposal	Define 4-6 evaluation metrics (e.g., feasibility score, latency impact, false positives): link each metric to a decision in the artefact: document assumptions.	Updated proposal evaluation section: metric table included clear justification in method.	NIST SP 800-82: IEC 6235: module notes: supervisor feedback if available.	Week 1
Improve Harvard referencing consistency and academic writing flow	Standardise in text citations: run a reference checklist; ensure every figure/table is captioned and referenced in text.	No missing references: consistent formatting: improved readability in unit summaries and evaluations.	Library guidance; UoEO study skills hub: reference manager.	Week 2
Develop stronger quantitative reporting habits	Complete two extra practice problems (hypothesis testing + effect size): write short method result interpretation notes.	Two worked examples in appendix: assumptions stated, interpretation avoids confusing statistical and practical significance.	Units 8-9 worksheets: lecture notes: practice datasets.	Week 2-3
Create an ethics and governance checklist for AI/OT use cases	Draft checklist covering transparency, access control, logging, data minimisation, retention, accountability: apply it to the proposal artefact.	Checklist included as an appendix: proposal ethics section updated with concrete controls and responsibilities.	ACM/BCS guidance: GDPR principles: module materials.	Week 3

Table 4. Professional Development Plan (PDP).

Assessment of Learning Outcomes and Evidence

The table below links module goals to proof in this portfolio and notes.

Learning outcome	**How I addressed it**	**Evidence**	**Where to find it**
Appraise professional, legal, social, cultural, and ethical issues affecting computing professionals	I analysed ethical and professional duties in my reflective activities and linked them to governance decisions in AI and security contexts.	Reflective Activity evidence: ethics discussion within reflection.	Digital Report: Units 1-2: Reflective piece: WHAT and SO WHAT
Appraise principles of academic investigation and apply them to a research topic	I refined a research topic into clear questions, aligned method choices with constraints, and justified my approach using research design principles.	Literature review plan: proposal framing and methodology notes.	Digital Report: Units 2-5: Evaluation sections 5-6
Evaluate critically literature, research design, and methodology, including data analysis processes	I evaluated strengths and gaps in my literature review and proposal design and demonstrated disciplined quantitative reasoning through hypothesis testing.	Literature Review evaluation: statistics worksheet outputs: proposal evaluation section.	Digital Report: Sections 4-6: Appendix A (stats)
Produce and evaluate critically a research proposal for the chosen topic	I produced a proposal with a defined artefact and evaluation approach, then reflected on improvements needed, including metrics and ethics details.	Proposal transcript and slides: proposal evaluation write up.	Digital Report: Section 6: Appendix C (proposal artefacts)

Table 5. Learning outcomes and evidence mapping.

Conclusion

My proof highlights gain in research design, scholarly work, stats thinking, and job reflection.

The top gain is trust in research choices that are backed up, recorded, and fit with ethics and real-world limits.

Figure A1: Trust boundary diagram for an illustrative smart grid / SCADA environment.

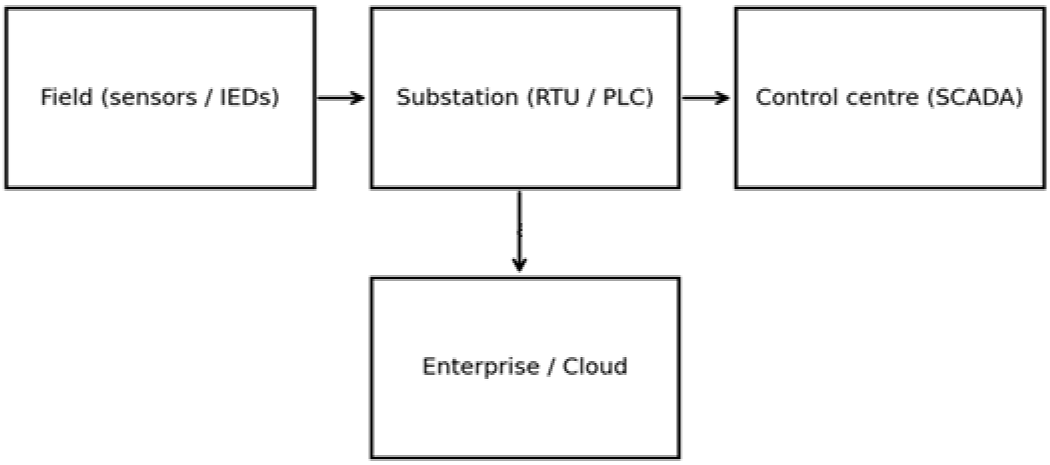

Figure A1. Trust boundary diagram.

Figure A2: Risk matrix used to summarise delivery risks in the research plan.

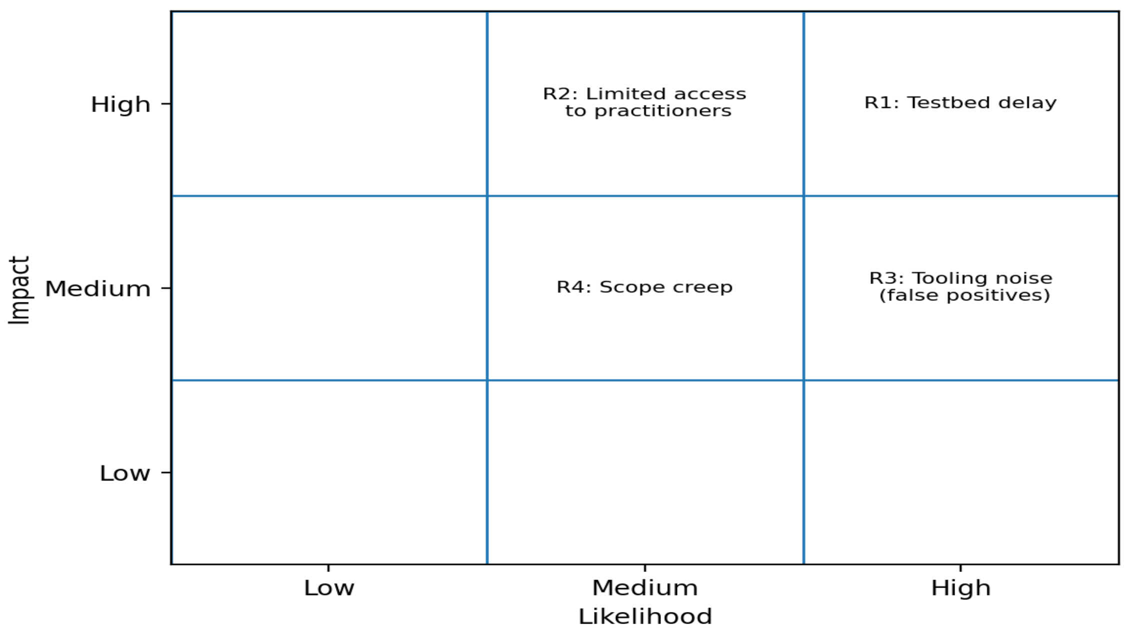

Figure A2. Risk matrix.

Figure A3: Histogram used to demonstrate distribution checking in quantitative analysis.

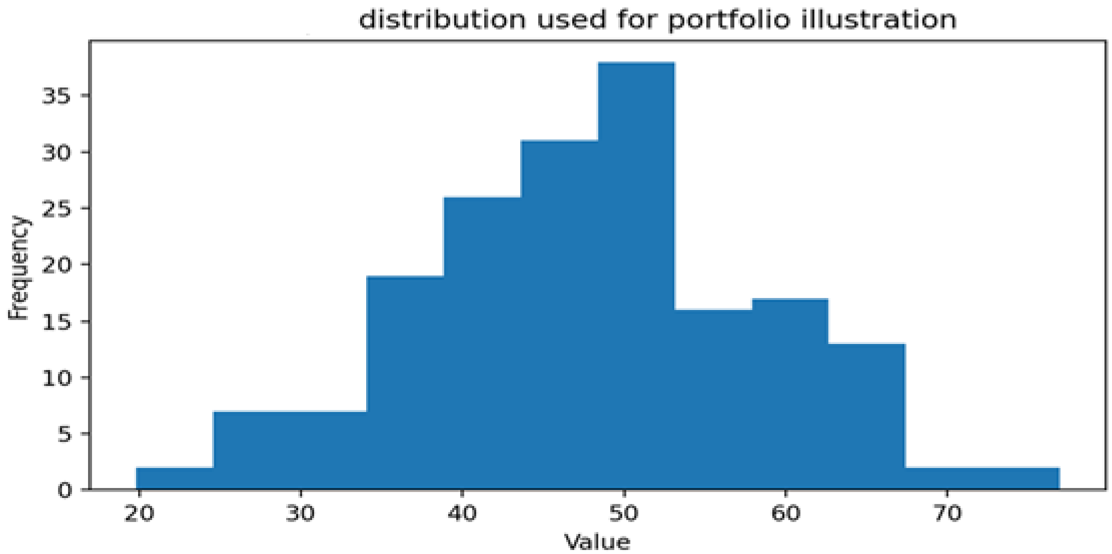

Figure A3. Histogram.

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Reflective Report

Learning Journey and Activities (WHAT)

At the module's start, I viewed research as basic school tasks: pick a topic, read papers, write a report. I knew research terms well. Yet I found it hard to shape them into solid choices. I also missed how work duties guide strong research in computing. This matters most when systems touch safety, reliability, and public trust.

Two tasks brought the lessons to life. First, my review of AI text tools and email safety linked tech fixes to legal and social effects. I once saw phishing as a tech battle of filters, rules, and scans. Attackers now craft real looking, tailored emails in bulk. This shifts the main issue: how can an organisation weigh safety steps against openness, fair auto choices, and blame when blocks halt real tasks (Floridi, 2019; Mittelstadt, 2019)?

Second, my work on IoT risks in smart grids and SCADA showed real world limits. In these systems, update times are short. Old gear cuts choices. Brief delays bring big risks. This guided my paper search and plan. The aim went beyond threat lists. It mapped risks to fixes that utilities can use without halting key services (NIST, 2015; IEC, 2018).

At the same time, stats tasks tested me another way. I handled basic stats with ease. But tests of ideas demanded care. I stated no change and change claims. I picked tests with reasons. I stuck to rules. I read results without overclaiming data support.

Challenges, Emotions, and Growth (SO WHAT)

My key shift in thought is this: good research needs a tight link from question to method, evidence, and claim. A weak spot makes it seem strong at first, but it falls apart on close check. I felt two common emotions along the way.

First came doubt. On the proposal, I felt pushed to cover all smart grid cyber security a broad field. This brought worry and stalled me. I fixed it by focusing on an artefact-led design science path: make a security tool kit, evaluate it in a set scene, then tweak based on clear results. This tied each question to a real output and check step (Creswell and Creswell, 2018).

Second was care. In ethics tasks and stats work, I grew wary of bold claims. For email security thoughts, it was easy to call auto filters a clear win. But they risk harm: biased blocks, bad reasons, slim appeal paths. This led me to frame it as rule choices, beyond just tech tweaks.

In stats drills, I saw the trap: a key stat result does not mean real world weight. Practice with hypothesis tests split stat proof from true impact. It taught me to share findings with care, and to spell out base ideas and limits. Sense only holds if the setup fits.

This change shaped how I checked my module work. In the lit review, I eyed source strength and proof for each point, not just echoes in texts. For the proposal, I went from listing methods to defending them against real limits like access ease, upkeep, and checks in live use.

I also saw my part in team tasks more clearly. With shared jobs, I had to show my outputs, progress shares, and feedback replies. This drove sharper writing and steady notes on choices, so others could track the logic with no extra help.

Skills Gained and Future Applications (NOW WHAT)

I shall use these lessons in four clear steps.

First, I will sharpen my evaluation definitions for future research. The proposal covered the testbed and attack tests. Now I will choose a few key metrics and explain each one. These cover latency effects, detection strength, false alarms, and a preset feasibility score. I will detail how to measure them and what tradeoffs pass muster.

Second, I will build ethics into design from the start. For auto filters or checks, I will plan clear logs, decision reasons, review access, and fix paths. This meets data rules for logs, emails, or system tracks.

Third, I will build strong stats skills with practice and clear reports. I aim to evaluate well and explain fully: aims, base ideas, methods picked, results shown, meanings drawn, and weak spots noted. I will write brief result notes that stay honest. These boosts trust in work and career.

Fourth, I will link thoughts to steps via my skills grid and SWOT map. After each big output, I will add one skill win and one gap to the grid. I will link proof shots to my online portfolio. I will check writing in stages: one for shape, one for sources, one for plain sense.

This module changed my view. I moved from just finishing jobs to making proof that others can check and use. Research and work now match state the issue sharply, pick sound tools, gather solid facts, share results to guide safe choices (Saunders, Lewis and Thornhill, 2019; Rolfe, Freshwater and Jasper, 2001).

Conclusion

This module changed my view of research. It turned a standard school task into a strict process that links real world issues, ethical duties, and choices based on facts. Direct work covered AI email protection, IoT threats to key systems, and data stats. From this, I saw good research needs clear links. The question must match the method, limits, proof, and results. I built a careful, ethical outlook too. I know spot limits in automation, risks of broad claims, and the need for openness and duty in computer setups. These lessons sharpened my skills to plan clear, doable, ethical studies. I share results with plain talk and care. In turn, I can create research and work that others trust, check, and use in elevated risk or sensitive areas.

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